

CUBICLE -08

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DAY 5 – TWO IN ONE DIGITAL OSCILLOSCOPE

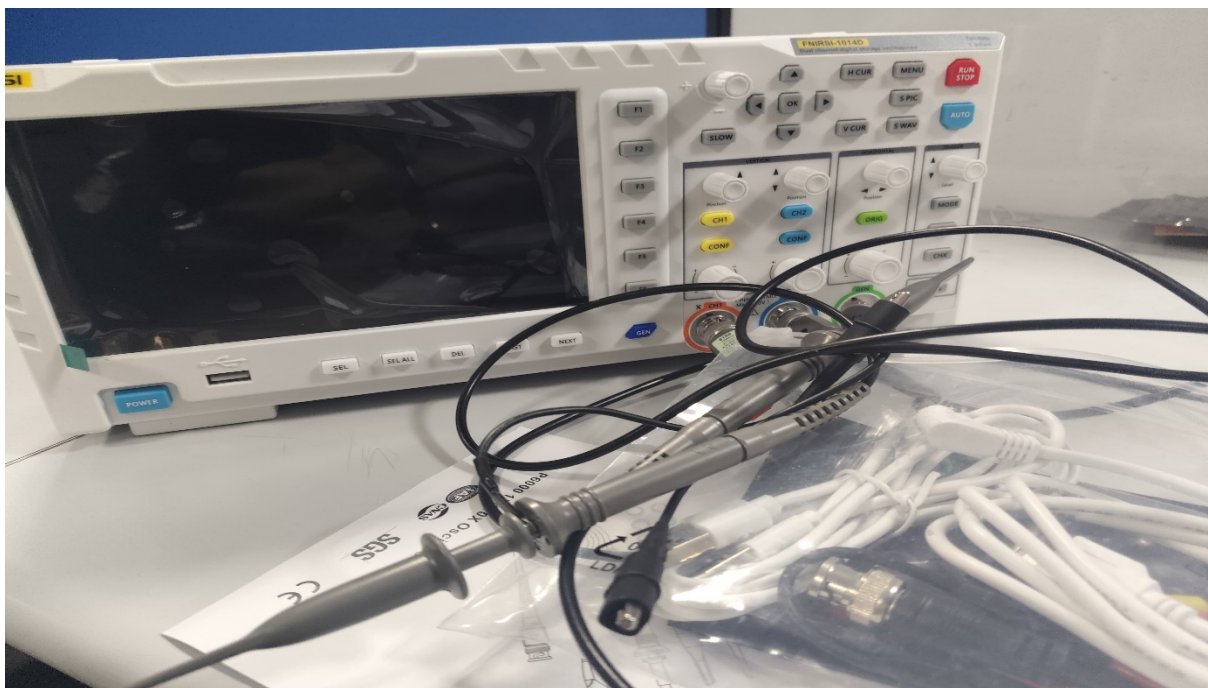
Date: 23-07-2025

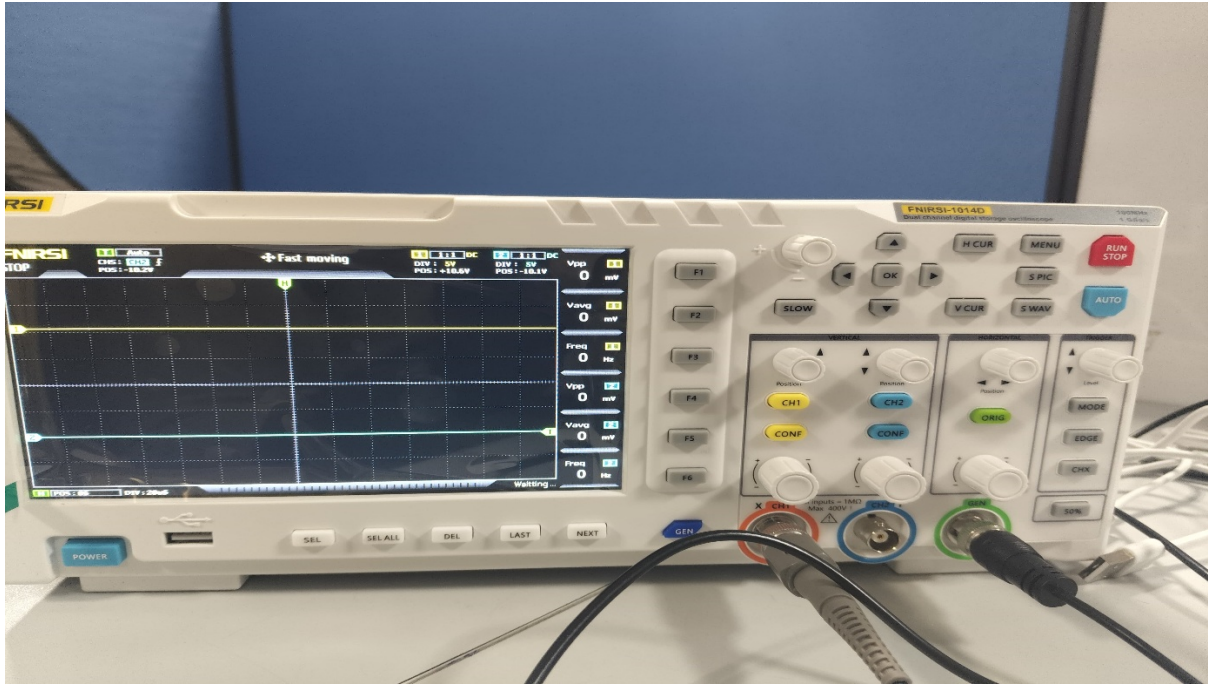
Domain: Embedded Systems and IoT

Topic: Two in One Digital Oscilloscope [Task 1 , Task 2]

Description of FNIRSI-1014D:

The FNIRSI-1014D is a dual-channel digital storage oscilloscope designed for beginners, hobbyists, and students. It offers reliable waveform analysis with a built-in signal generator, compact size, and a budget-friendly price. It is commonly used for electronics debugging, Arduino projects, and classroom experiments.





Task 1: PWM Signal Analysis

Objective:

To visualize and analyze the PWM waveform on an oscilloscope with different duty cycles using an Arduino.

Components Required:

- Arduino Uno (or compatible board)
- USB Cable
- Oscilloscope (e.g., FNIRSI-1014D)
- Oscilloscope Probes
- Jumper Wires

Arduino Code:

```
void setup() {  
  pinMode(9, OUTPUT); // Set Pin 9 as output  
}
```

```
void loop() {  
  analogWrite(9, 64); // 25% duty cycle  
  delay(2000);  
  
  analogWrite(9, 128); // 50% duty cycle  
  delay(2000);  
  
  analogWrite(9, 192); // 75% duty cycle  
  delay(2000);  
}
```

Code Explanation:

- Pin 9 is configured as PWM output.
- The analogWrite() function sets PWM signal:
 - 64 → 25% duty cycle (64/255)
 - 128 → 50% duty cycle
 - 192 → 75% duty cycle
- Every duty cycle is held for 2 seconds before changing.

Connection Setup:

Connect the following:

- Oscilloscope CH1 probe tip → Arduino Pin 9
- Oscilloscope Ground clip → Arduino GND (Ground)

Oscilloscope Settings:

- Time/Div: 0.5 ms/div (or 1 ms/div)
- Volt/Div: 2 V/div
- Coupling: DC

- Trigger Mode: Auto or Normal
- Trigger Source: CH1

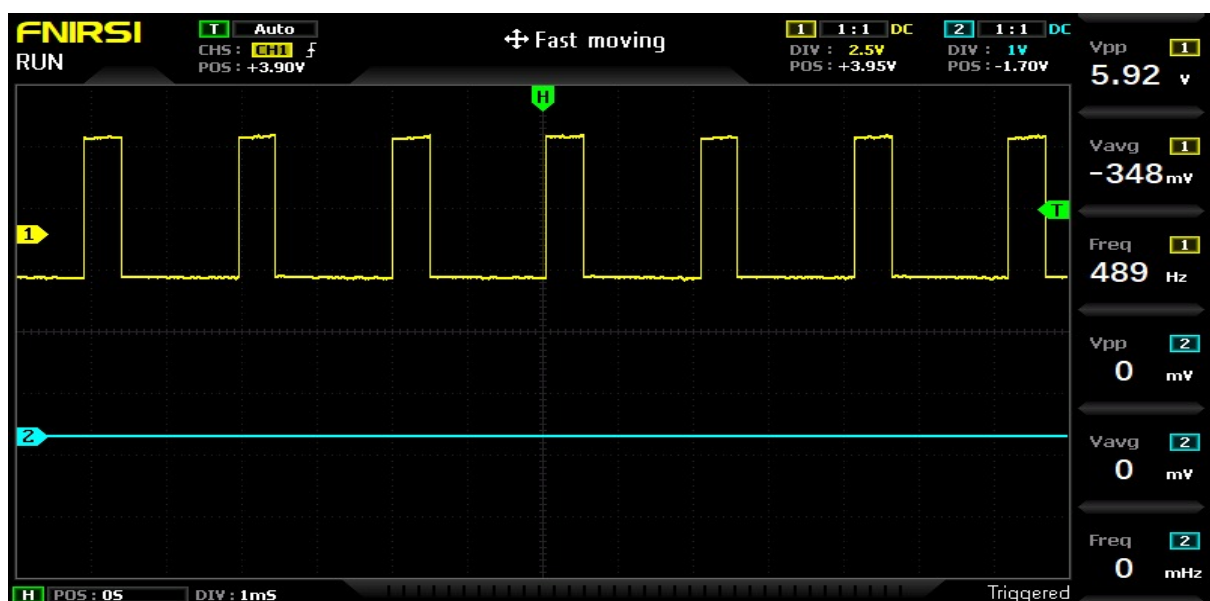
Observation:

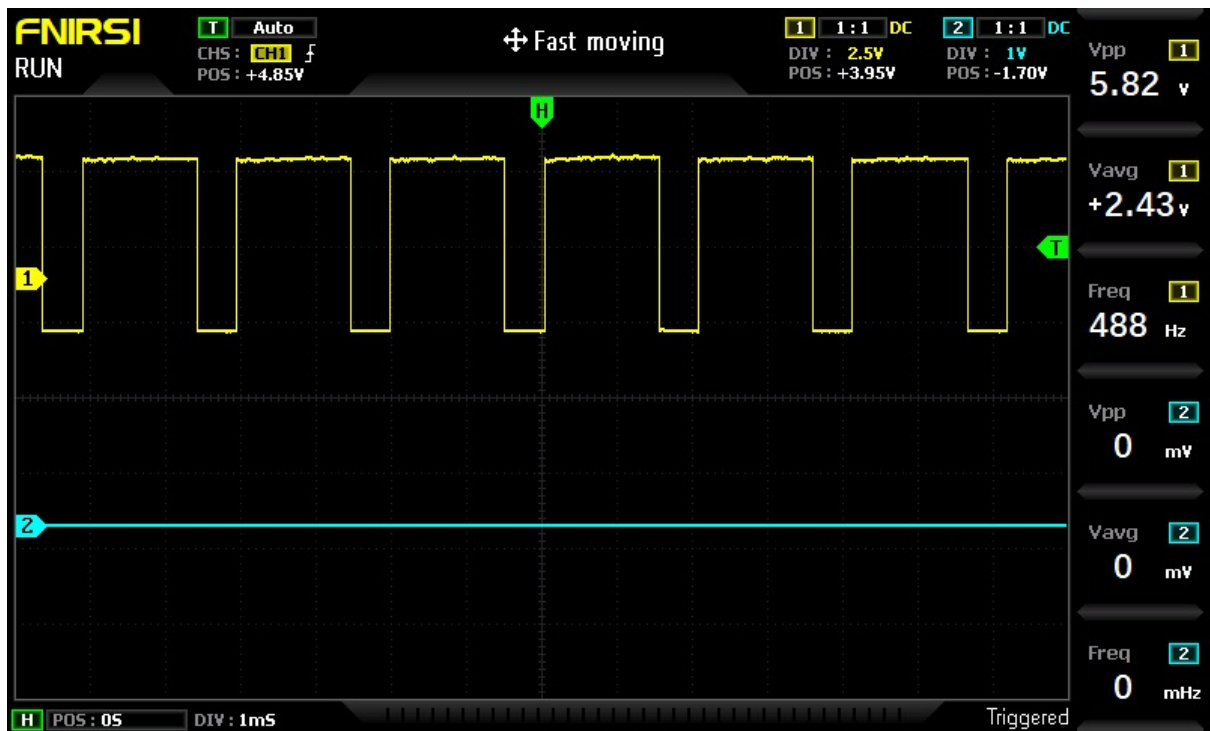
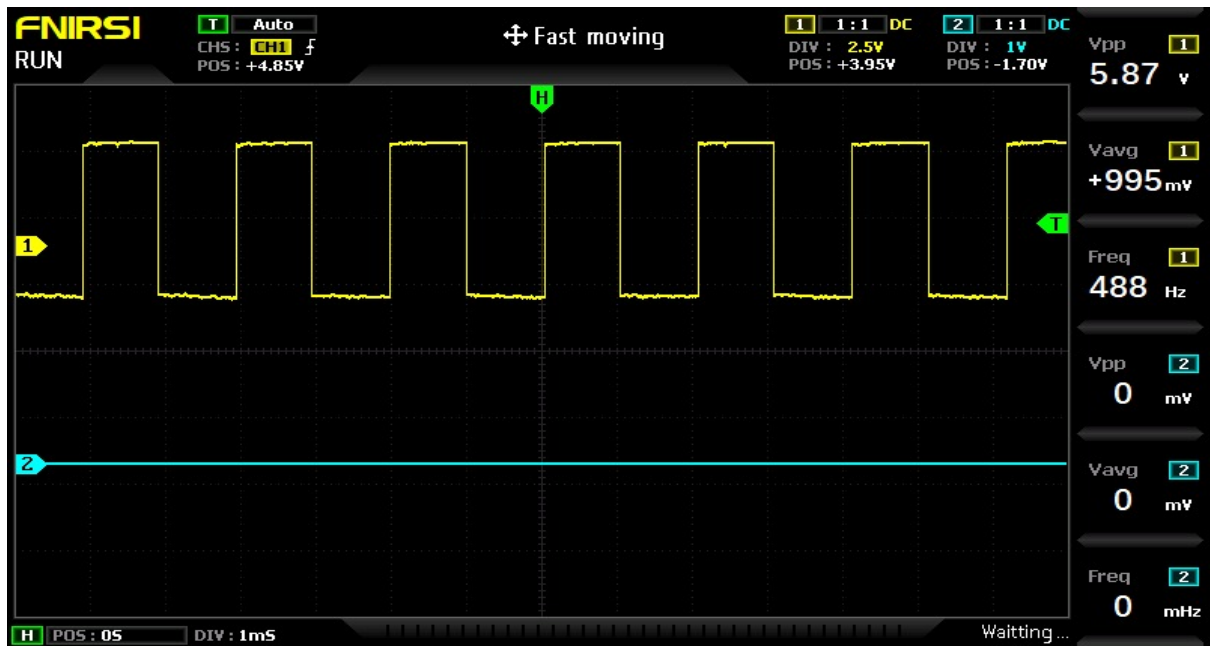
- A square waveform will be displayed on the oscilloscope.
- Frequency remains constant (~ 490 Hz).
- Duty cycle changes every 2 seconds:
 - 25% Duty: Thin Pulse
 - 50% Duty: Medium Pulse
 - 75% Duty: Wide Pulse

Expected Output:

- Stable square waveform with constant frequency.
- Duty cycle changes clearly every 2 seconds.
- Example:
Thin - Medium - Wide - Repeat

Screenshots:





Task 2: Measure Frequency from Arduino

Objective:

To manually generate a square wave using Arduino and measure its frequency, period, and voltage levels using a Digital Storage Oscilloscope (DSO).

Components Required:

- Arduino Uno
- USB Cable
- FNIRSI-1014D Oscilloscope (or similar)
- Jumper Wires

Arduino Code:

```
int wavePin = 9;
```

```
void setup() {  
  pinMode(wavePin, OUTPUT);  
}
```

```
void loop() {  
  digitalWrite(wavePin, HIGH);  
  delayMicroseconds(998); // 1 ms ON  
  digitalWrite(wavePin, LOW);  
  delayMicroseconds(998); // 1 ms OFF  
}
```

Code Explanation:

- Pin 9 is configured as an output.

- The loop alternates the pin between HIGH and LOW for 1 millisecond each.
- This produces a square wave with:
 - 1 ms HIGH
 - 1 ms LOW
- The total period is 2 ms, which results in a frequency of approximately 500 Hz.

Connection Setup:

- Connect Arduino Pin 9 to Oscilloscope Channel 1 probe.
- Connect Oscilloscope Ground to Arduino GND.

Oscilloscope Settings:

- Time/div: 0.5 ms/div
- Volt/div: 2 V/div
- Use the cursor or auto measurement tool on the DSO to:
 - Measure frequency (should be approximately 500 Hz)
 - Measure period (2 ms total: 1 ms HIGH + 1 ms LOW)
 - Confirm voltage levels (LOW = 0V, HIGH = ~5V)

Observation:

- A clean and symmetrical square wave is displayed on the oscilloscope.
- The wave shows equal HIGH and LOW durations.
- The measured period is close to 2 ms.
- The measured frequency is around 500 Hz.
- Voltage levels alternate between 0V and approximately 5V.

Expected Output:

- Square wave with:

- Equal HIGH and LOW times (symmetrical waveform)
- Frequency around 500 Hz
- Voltage levels: LOW $\approx 0\text{V}$, HIGH $\approx 5\text{V}$
- Stable and clear waveform observed on the oscilloscope.

Screenshots:

